Searching Techniques: Linear Search & Binary Search Data Structures

> **Definition: Searching** is the algorithmic process of locating the position of a target key element `K` within a collection of items (such as an array). The two primary searching techniques are **Linear Search** and **Binary Search**.

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1. Detailed Technical Explanation

1. Linear Search (Sequential Search)

- **Concept:** Sequentially inspects every element in the array from index `0` to `N - 1` until the target element is found or the end of array is reached.

- **R

```c
int line
```

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2. Binary Search (Divide and Conquer)

- **Concept:** Repeatedly divides the search interval in half by comparing the target key with the middle element `arr[mid]`:

- If `a

```c
// Itera
```

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2. Comparison: Linear Search vs Binary Search

| Feature | Linear Search | Binary Search |

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3. Quick Recall Flow

```

Linear
```